

# LWIR Camera Calibration

A measured lens, a known mounting direction, and evidence you can keep. Work through the Tiny1-C calibration one clear checkpoint at a time.

## ON THIS PAGE

- |                                          |                                        |
|------------------------------------------|----------------------------------------|
| 1. Know What You Are Calibrating         | 8. Check Distances Before Checking GPS |
| 2. Gather Ordinary Materials             | 9. Install The Profile Deliberately    |
| 3. Make A Target The Camera Can See      | 10. Keep Temperature Checks Separate   |
| 4. Freeze The Camera Setup               | 11. The Flight Accuracy Gate           |
| 5. Take A Useful Set Of Photos           | 12. Quick Troubleshooting              |
| 6. Fit The Lens On The Laptop            | 13. Save A Reusable Calibration Pack   |
| 7. Measure The Mounting Without Guessing | References And Scope                   |

A workshop guide for the Tiny1-C, CATIA and a fixed, downward-looking camera.

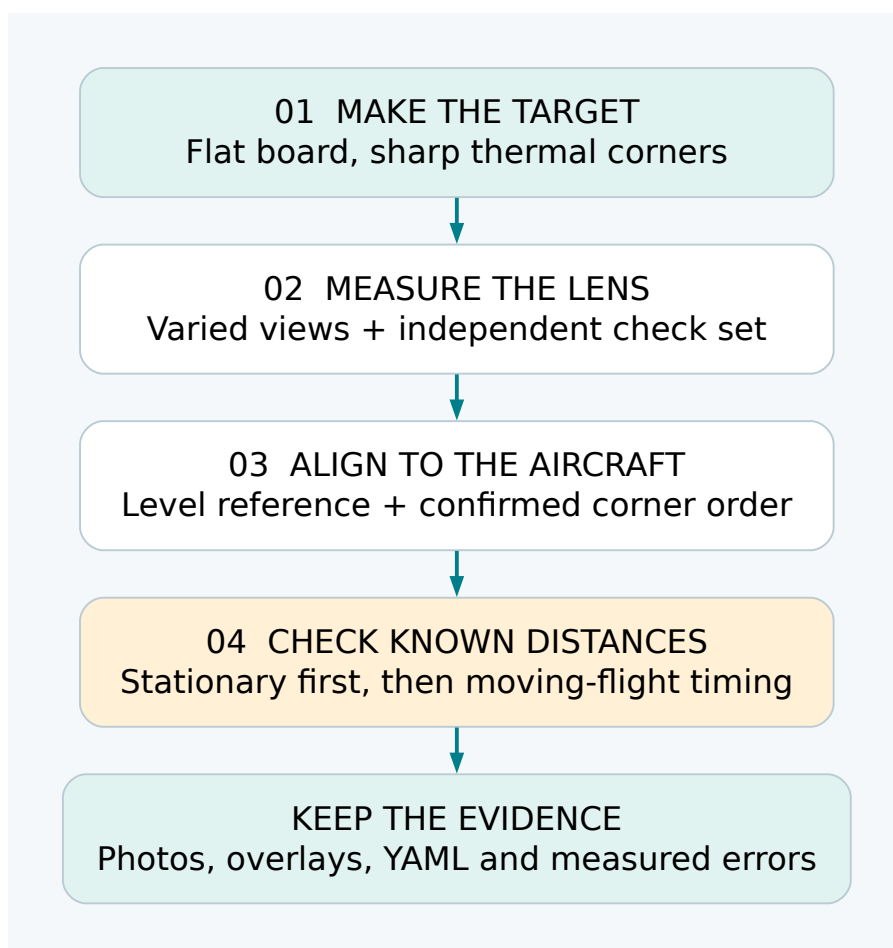
[Open the illustrated HTML guide](#) | [Back to CATIA](#)

**The goal:** turn a hotspot pixel into a trustworthy camera direction. You can measure the lens and mounting with a flat homemade target, a tape measure, a spirit level and a computer. Accurate flight GPS also needs exposure-time position and attitude; calibration does not remove that requirement.

# 1. Know What You Are Calibrating

There are three different jobs. Do them in this order so that a bad temperature reading is not mistaken for a bad lens, or a timing error for a mounting error.

| Job                | What it fixes                                       | What you will keep                                           |
|--------------------|-----------------------------------------------------|--------------------------------------------------------------|
| Lens geometry      | Pixel direction, lens distortion and optical center | Measured intrinsics and distortion in a YAML file            |
| Mounting alignment | The camera's direction relative to the aircraft     | A measured <code>camera_to_body</code> rotation in that file |
| Temperature check  | Whether reported surface temperatures are plausible | A separate check record, not lens coefficients               |



Allow an unhurried workshop session, then a separate outdoor check. You do not need RTK or a blackbody to measure lens geometry. You do need a traceable temperature reference to claim calibrated absolute temperature, and a surveyed reference to prove absolute GPS accuracy better than ordinary GNSS.

**Finish line for this guide:** a measured lens profile, a measured mounting profile, saved evidence, and a short list of any checks still outstanding. No file will be automatically marked verified.

## 2. Gather Ordinary Materials

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- The actual camera, lens and final mounting bracket.
- A rigid flat board, preferably a smooth metal sheet with safe, covered edges.
- Thin matte tape and smooth aluminum foil tape, a ruler, square and craft knife.
- A spirit level, tape measure and string with a small plumb weight.
- Gentle warmth: a warm room, warm water container behind the board, or a low-temperature heating pad used according to its instructions.
- A laptop and a safe support for the camera or aircraft.
- For temperature checks: a reasonably calibrated contact probe and a matte high-emissivity target surface whose temperature can be measured.

**Remove the propeller and disconnect propulsion power.** Secure the aircraft; do not balance it on loose objects. Keep liquids away from electronics. Do not use fire, boiling water, a heat gun or a hot plate to make a calibration target. Never point the LWIR camera at the sun or a powerful hot source.

An ordinary black-and-white printed chessboard often looks nearly uniform to LWIR. Visible ink color is not enough. The camera needs a difference in thermal emission or reflected long-wave infrared radiation.

### 3. Make A Target The Camera Can See

#### Build one small test patch first

Put one matte square beside one smooth foil square on the same flat backing. View it with the LWIR camera, from the intended distance and a modest tilt. Try gentle, uniform warming if the squares merge together.

**Continue only when the boundary is sharp and stable.** Foil reflects the thermal surroundings: your body, a radiator or the sky can change its apparent brightness. Move yourself and warm objects away from the reflection. Neither the displayed brightness nor the apparent foil temperature is a temperature reference. A contrast reversal is acceptable if all corners remain clear.

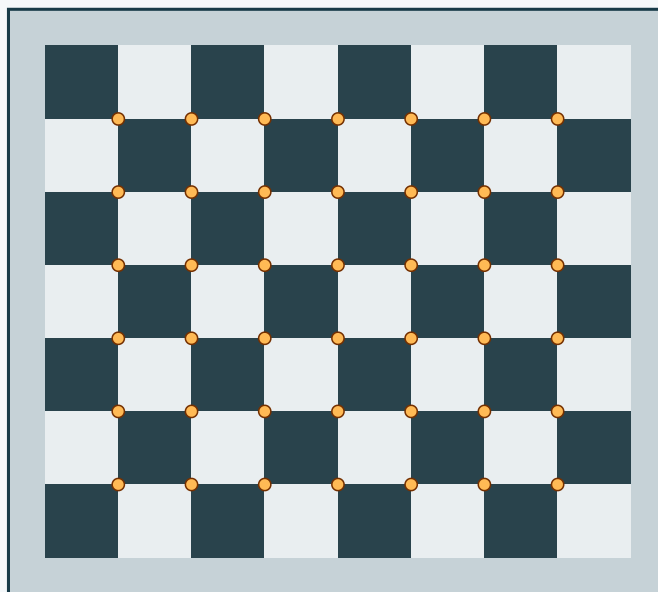
If the patch only works from one angle, improve it before building the full board. Wrinkled foil, thick tape, bowed backing and local heating make fitting needlessly difficult. A purpose-made thermal checkerboard is the fallback if the homemade board cannot give repeatable edges.

#### Build the full checkerboard

Use **8 columns by 7 rows of squares**. That makes **7 by 6 inner corners**, which is what the helper asks for. Add a uniform border at least half a square wide. Cut clean edges; avoid overlaps and gaps at the crossings.

#### Build squares. Count inner corners.

Construction schematic / dimensions come from your ruler



**8 x 7 squares**

**7 x 6 inner corners**

■ Thin matte tape

□ Smooth foil

Dots show crossings.

Do not add dots to the target.

Flat backing, clean edges,  
uniform border, no overlaps.

5 pitches = 250 mm gives 50 mm per square

Example only. Measure both directions on the finished board.

*Construction diagram, not an image captured by the camera. The colored dots mark the internal corners; they are not objects to put on the board.*

Measure the finished square **pitch**, from one grid line to the next, not the width of the remaining exposed foil. Measure across several squares and divide by their count. Check both directions. For example, five pitches measuring 250 mm means 50 mm per square. If horizontal and vertical pitches differ or the grid is visibly crooked, rebuild it rather than averaging the error away.

Do not print the illustration and assume its physical scale is correct.

## Choose a size that stays in focus

Start by finding a distance where the final flight-focus setting gives sharp corners. **Do not refocus just to make a small indoor board usable.** A change in focus can change the calibration; use a larger board farther away instead.

For planning only, square width in pixels is approximately:

```
square_pixels = focal_length_pixels * square_pitch_metres / distance_metres
```

With the unverified 4.3 mm / 12 micrometre assumption, focal length is about 358 pixels. A 50 mm square at 1 m spans about 18 pixels. Eight squares span about 143 pixels before the border. At 3 m, a 150 mm square gives the same angular size, but needs a much bigger board. These are sizing examples, not measured Tiny1-C specifications or instructions to change focus.

Aim for roughly 10-20 pixels per square in many of the photos; avoid tiny, blurred squares. For a longer lens, increase distance or reduce target size while keeping the target sharp and fully visible.

**Checkpoint:** the complete grid and border fit in the image; internal crossings are recognizable at 100% view, including moderate tilts.

## 4. Freeze The Camera Setup

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1. Choose the lens that will fly. Make separate profiles for 4.3 mm and 9.1 mm.
2. Set flight focus and secure it without moving the lens.
3. Use the final camera orientation, resolution and capture settings.
4. Let the camera settle according to its manual. Wait for automatic shutter correction to finish before keeping a frame.
5. Save native JPEGs. Do not rotate, crop, resize, undistort or annotate them.

Normal Tiny1-C combined capture has a 256x192 visible image and an aligned 256x192 temperature plane. Check the actual saved files. The 256x191 mock fixture is not a reason to crop a real 256x192 capture.

On MORA, with CATIA stopped so it is not using the camera, an existing deployed LWIRcam can capture a bench photo:

```
cd /home/air/digital_cam
mkdir -p calibration/photos
./lwircam --capture --bare --output calibration/photos/view01.jpg
```

Take the next photo with a new filename. `--bare` is a compatibility option; current capture remains unfiltered. Single-shot capture may take time to warm up. Do not start another instance before it finishes, and do not use `--mock-image` or `--mocktransform` for real calibration photos.

Copy the finished JPEGs and their `.jpg.raw` companions to the laptop using your normal file-transfer method. The helper needs only the JPEGs; retain the native combined frames for subsequent temperature analysis. Keep matching filenames and do not resize either part. This guide does not require ExifTool on MORA.

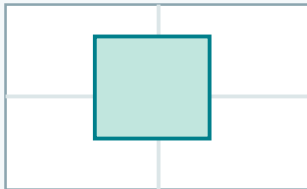
## 5. Take A Useful Set Of Photos

Keep the camera still and move the target for this stage. Secure the target at each pose; do not take photos while it is moving.

### Let the corners visit the whole image.

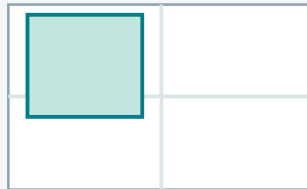
Keep the complete border visible / keep flight focus unchanged

**01 Center**



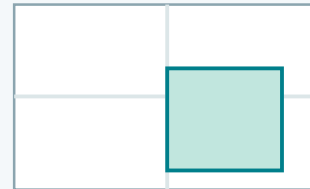
Target footprint in sensor

**02 Upper left**



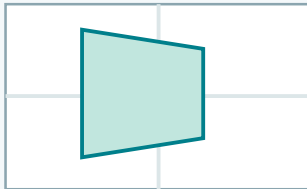
Target footprint in sensor

**03 Lower right**



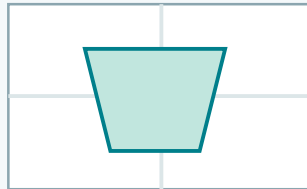
Target footprint in sensor

**04 Left/right tilt**



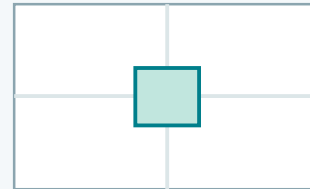
Target footprint in sensor

**05 Forward/back tilt**



Target footprint in sensor

**06 Change distance**



Target footprint in sensor

### 25-35 varied fitting views

#### + 5-8 new views kept only for checking

Include all edges and corners; these six poses are examples, not the full set.

*Schematic positions, not prescribed rotations. Keep the entire board visible and its corners sharp. Edge views matter as much as the centered view.*

Collect about **25-35 usable views**, with variety rather than a burst of nearly identical frames:

- Several central views, including one almost front-on.
- Views near each image corner and each edge, with the full target visible.
- Moderate left/right tilt and forward/back tilt, roughly 15-35 degrees if contrast and focus permit.
- A few changes in distance and in-plane rotation.

Set aside **5-8 additional views** as a check set. They must be genuinely new views, not duplicates of fitting photos. Do not use them to fit the lens.

Do not pursue a target number at the expense of quality. A frame with clipped corners, glare, a shutter event or motion blur should be retaken. The helper skips frames where it cannot find the full pattern and prints their filenames.

**Checkpoint:** the corners have visited most of the sensor, including the edges, and the board is not parallel to the camera in every picture.

## 6. Fit The Lens On The Laptop

### Prepare the desktop helper

The accompanying [calibration helper](#) uses OpenCV. It runs only on the development computer; Python, NumPy and desktop OpenCV are not new flight-application dependencies. Its output matches CATIA's YAML keys.

From the Paparazzi repository root:

```
python3 -m venv .venv-lwir-calibration
.venv-lwir-calibration/bin/pip install numpy opencv-python-headless
```

Keep fitting images in `calibration/photos/`, independent check images in `calibration/check/`, and mounting images in `calibration/mount/`. These paths are examples relative to your current directory. Do not put annotated previews back into the input directories.

### Run the fit

For a board with 7x6 internal corners and measured 50 mm pitch:

```
.venv-lwir-calibration/bin/python \
  sw/airborne/modules/digital_cam/catia/documentation/lwir_calibrate.py lens \
  --images calibration/photos --check calibration/check \
  --columns 7 --rows 6 --square-mm 50 \
  --output calibration/tiny1c-lens.yml
```

Use your measured pitch. The helper requires at least 15 detected fitting views, rejects mixed image sizes and refuses to overwrite an existing result. Choose a new output name for another attempt so earlier evidence stays intact. It writes:

- The lens YAML, with a **nominal, unmeasured mounting** and `verified: 0`.
- A report giving detected/skipped images, per-view errors and check-set errors.
- Enlarged corner previews for inspection. Enlarging these previews does not change the calibration images or fitted pixel coordinates.

Open the preview images first. The colored corners must sit on the true crossings all the way around the grid. A small reported error does not rescue incorrect corner detection.

### Decide whether to keep the result

Reprojection RMS is the typical discrepancy, in pixels, between the detected corners and the fitted model. It is **not GPS error in metres**. The check set uses fixed lens parameters and fits only each board pose, so it is a useful consistency check, not an independent survey of absolute accuracy.

As workshop triage, not camera specifications:

| Result                                                                   | Next action                                             |
|--------------------------------------------------------------------------|---------------------------------------------------------|
| Fit and check errors mostly below about 0.5 pixel, with correct overlays | Continue to mounting and independent physical checks    |
| Some views around 0.5-1 pixel                                            | Inspect those frames, grid flatness and edge coverage   |
| Many errors above 1 pixel or the check set much worse than fitting       | Fix the target, focus or pose variety before continuing |

Do not delete difficult images just to improve the score. Remove only images with an identifiable problem, replace them with good views of the same region, and save the reason. A model fitted only near the image center is not validated at the edge.

Compare two independently collected sets. Large changes in focal length, optical center or predicted edge rays mean the setup needs attention. There is no universal coefficient threshold that guarantees a good calibration.

**Checkpoint:** you have inspected every retained overlay and the independent check views behave comparably to the fitting set.

## 7. Measure The Mounting Without Guessing

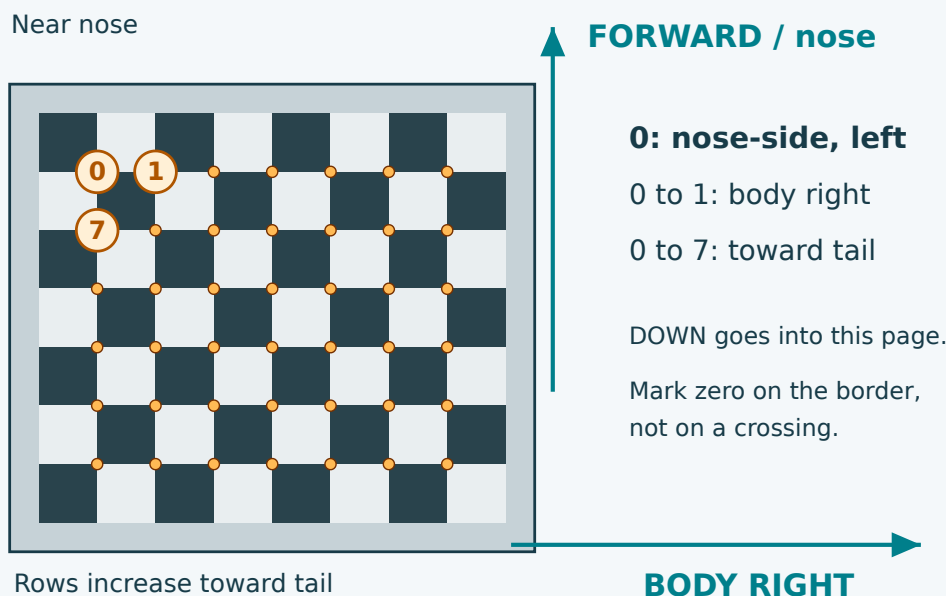
Lens calibration does not tell CATIA which way the camera points relative to the aircraft. Near-zero GPS-to-camera distance removes a translation concern; it does not remove a small angular mounting error.

### Make the board a body-axis reference

Remove the propeller. Secure the complete aircraft on a support above the flat target, with the camera in its final bracket. Do not move that bracket between lens fitting, mounting measurement and flight.

#### Give the board the aircraft's axes.

Top view / board and aircraft body reference must be level



#### Confirm labels on the saved preview before trusting the rotation.

This ordering applies to the 7-column inner-corner pattern shown here.

1. Establish the aircraft body reference using the flight controller's defined level attitude and correctly configured board alignment. Check the actual airframe datum with a level; a convenient sloping fuselage surface is not automatically that datum. If these disagree, resolve the FC alignment first.
2. Level the target in the same horizontal reference. The aircraft body axes and the board reference must agree; the helper cannot correct a tilted stand.
3. Align a board column direction with **aircraft right**. Increasing row number must run **toward the tail**. Use string, a square and the aircraft centerline, not a phone compass beside electronics.
4. Put the board on a stable support where the full pattern is sharp. It may be offset from directly under the camera; exact height and centering are not needed for the rotation fit. Keep it horizontal and aligned to the aircraft.
5. Mark the board's near-nose, aircraft-left inner corner on its border as the intended corner zero. Do not cover an actual checker intersection.

Misleveling by one degree is already approximately 0.70 m at 40 m AGL near nadir. A phone level may help rough setup but is not proof of a precise datum.

## Fit and inspect one mounting view

Capture a photo without disturbing the alignment, then run:

```
.venv-lwir-calibration/bin/python \  
  sw/airborne/modules/digital_cam/catia/documentation/lwir_calibrate.py mount \  
  --image calibration/mount/mount01.jpg \  
  --lens calibration/tiny1c-lens.yml \  
  --columns 7 --rows 6 --square-mm 50 \  
  --output calibration/tiny1c-mounted.yml
```

Inspect the numbered preview. **0 must be the near-nose, aircraft-left inner corner; 1 must be the next inner corner toward aircraft right; 7 must be the first corner in the next row toward the tail** for a 7-column pattern.

A checkerboard can be found in the opposite order. If it is reversed by 180 degrees, rerun with `--reverse-corners` and a new output name. If it is rotated sideways, or the order does not match the board reference, stop and correct the setup; do not try random signs in the matrix. Lens fitting does not need this absolute corner orientation, but mounting measurement does.

The helper uses measured lens parameters and OpenCV `solvePnP`, with board points expressed in body-aligned axes. It exports the inverse rotation as `camera_to_body`. Translation is fitted only to locate the board; it does not become a GPS lever arm. The mounting YAML still has `verified: 0`.

Repeat at least three times, repositioning the board while keeping its axes aligned and its plane level. Compare the resulting rotations and ground-check errors. Do not average matrix entries by hand. Nearly front-on planar pose estimates can be sensitive to corner noise: a low reprojection error alone does not settle their angular accuracy. If repeated results disagree beyond your error budget, improve the target size/reference setup or have the mounting measured with a better fixture. Keep the best-supported result, not merely the one closest to the nominal matrix.

**Checkpoint:** corner order is physically confirmed, independent mounting measurements agree, and the next ground check supports that alignment.

## 8. Check Distances Before Checking GPS

First test the projection in a stationary, local setup. This separates camera geometry from ordinary GNSS errors and camera-trigger delay.

1. Secure the aircraft above a level patch with propulsion disabled. Measure the camera height above that patch and use a plumb line to mark its ground projection. Keep supports out of the image.
2. Place a few small, safely warmed matte targets at measured offsets: center, forward, right, and near the usable image edges. Mark their centers and measure offsets with a tape. They must be large enough to occupy multiple pixels, well separated, and within the detector's operating range.
3. Use correctly tagged stationary shots with actual position, attitude and AGL. Do not substitute a stale in-flight pose or the simulated 40 m EXIF.
4. Check direction first: a target to aircraft right must not project left. Check all four directions before interpreting small distance differences.
5. Repeat with modest supported bank/pitch angles and changed heading, keeping the actual pose and height known. The target coordinate should remain fixed within the uncertainty of your measurements.

**Practical limitation:** the production detector normally needs a sufficiently hot target (generic minimum about 60 C, and stronger tray criteria). Do not heat a household target dangerously just to make it pass. Lens and mounting fitting use checker corners and do not require hotspot detection. A geometric check can instead use recorded corner/target pixels and the same ray projection, with help from the developer. There is currently no end-user arbitrary-pixel CLI.

For hot-target end-to-end checks, use a suitable controlled target and safe handling procedure. The ordinary checkerboard is not automatically a fire-test target. CATIA may report zero hotspots on it even when lens calibration is good.

Tape measurements validate relative geometry, not absolute latitude/longitude. A non-RTK M10N fix cannot establish a sub-metre surveyed truth point. For an absolute test, compare against an independently surveyed target, and record the uncertainty of both reference and aircraft position.

### Read error patterns rather than chasing numbers

| Pattern                                                      | Likely place to check first                         |
|--------------------------------------------------------------|-----------------------------------------------------|
| Error grows toward image edges                               | Lens fit, distortion, mismatched image size or crop |
| Similar offset everywhere that rotates with aircraft heading | Camera mounting alignment                           |
| Offset scales with height                                    | Angular alignment or focal-length error             |
| Error changes with speed or turn direction                   | Exposure/pose timing or motion blur                 |
| All targets share a slowly drifting displacement             | GNSS bias                                           |
| Error changes over sloping ground                            | AGL reference and flat-ground assumption            |

These are diagnostic clues, not unique diagnoses. Check one cause at a time.

## 9. Install The Profile Deliberately

Keep the lens-only file and reports unchanged. In the final mounting profile, change `verified: 0` to `verified: 1` **only after** you have reviewed lens evidence, confirmed corner order and mounting axes, and passed the independent geometry checks. This flag means you reviewed calibration; it does not mean GPS timing or field accuracy is certified.

The runtime expects these keys:

| Key                                                                                     | Meaning                                                                             |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <code>image_width</code> ,<br><code>image_height</code>                                 | Exact native capture dimensions                                                     |
| <code>fx</code> , <code>fy</code> , <code>cx</code> , <code>cy</code>                   | Focal lengths and optical center in original image pixels                           |
| <code>k1</code> , <code>k2</code> , <code>p1</code> , <code>p2</code> , <code>k3</code> | Brown-Conrady distortion, in that order                                             |
| <code>camera_to_body</code>                                                             | Nine row-major entries mapping camera right/down/forward to body forward/right/down |
| <code>verified</code>                                                                   | Your explicit calibration-review gate, 0 or 1                                       |

Use the original fitted camera matrix, **not** a cropped or optimized display matrix from an undistortion preview. Do not undistort saved images first: LWIRcam already undistorts the ray during projection.

Transfer the reviewed YAML to MORA, for example `/home/air/digital_cam/calibration/tiny1c-mounted.yml` , and set this environment variable in the CATIA service's actual startup configuration:

```
LWIRCAM_CALIBRATION=/home/air/digital_cam/calibration/tiny1c-mounted.yml
```

An `export` in your interactive SSH terminal does not update a service that is already running. Restart the relevant service through your normal deployment procedure while the aircraft is safe on the ground. This guide does not assume a particular service-unit name.

To analyze a copy of an existing flight-tagged JPEG on MORA:

```
cd /home/air/digital_cam
cp photos/l000001.jpg calibration/check-shot.jpg
cp photos/l000001.jpg.raw calibration/check-shot.jpg.raw
./lwircam --geolocate calibration/check-shot.jpg \
  --calibration calibration/tiny1c-mounted.yml
```

Keep this JPEG/raw pair together. The raw companion is the original combined Tiny1-C frame, not another JPEG. The command updates the JPEG in place and leaves the raw bytes untouched. Camera GPS remains the camera position; target GPS and center temperature go in EXIF UserComment under `LWIR_HOTSPOTS_V1` . Missing or unverified real-camera calibration suppresses coordinates.

Optional inspection on the development PC:

```
exiftool -b -UserComment calibration/check-shot.jpg
```

**ExifTool is not required by the final application.** CATIA and LWIRcam use bundled libexif. The helper and inspection commands are workshop tools only.

# 10. Keep Temperature Checks Separate

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The checkerboard measures geometry. Its foil squares deliberately exploit reflection and must not be used to calibrate temperature.

For a basic sanity check, use a large, uniform matte surface at a safe stable temperature, with a contact probe attached to the surface being observed. Make the observed patch large enough to cover many pixels. Let the probe and surface settle; check for gradients and avoid reflections of your hands. Compare several readings at two or three safe temperatures, including ambient and a gently warmed surface. Record distance, ambient conditions, surface material, emissivity settings and probe uncertainty.

The temperature of water behind a plate is not automatically the temperature of its front surface. A kitchen thermometer in that water is not a reference for a tiny remote pixel. Shiny metal can look hot or cold because of reflected radiation, even when a contact probe gives a different reading.

This exercise can reveal a gross problem. It cannot prove accuracy at fire temperatures, validate saturated pixels or replace a traceable blackbody calibration. Do not write vendor factory correction tables from this check. For quantitative high-temperature work, use an appropriate calibrated reference and the vendor procedure. Runtime values remain sensor-apparent temperatures, not automatically emissivity/atmosphere-corrected true surface temperatures.

**Checkpoint:** you know what your temperature check establishes, and what it does not. Keep that record with the geometric calibration, not inside its lens coefficients.

# 11. The Flight Accuracy Gate

---

The capture server now tracks frame stability continuously, including between requests. It no longer starts an eight-frame wait for every shot: an already stable stream can use the next acceptable frame acquired after the request. The startup warm-up and eight-frame recovery criterion remain after instability. These are image-content heuristics, not a Tiny1-C exposure-ready signal. Eight has not been established as the minimum required by hardware. Motion, texture, or a shutter correction can still trigger recovery; verify this on the camera before reducing or removing those remaining checks.

The server measures each request-to-selected-frame **arrival** interval using a monotonic clock at SDK callback entry, before application copying, validation, JPEG encoding, raw writing and analysis. Each callback delivery can be used only once; sequence gaps reset stability. This avoids a reproduced SDK polling issue where the same latest frame could be returned repeatedly after processing pauses. CATIA records `request_to_frame_arrival_s` in EXIF. This is not yet the interval from the FC pose timestamp to exposure: transport, scheduling before server acceptance and sensor/USB buffering remain unmeasured.

The Tiny1-C runs continuously; requesting a still does not start an exposure. At 25 Hz, the 40 ms frame period does not tell us integration time. Available SDK evidence did not confirm variable integration time or a numerical fixed lag. We should not assume visible-camera auto-exposure behaviour. A fixed effective lag is a reasonable model to test, not a proven camera constant.

## Optional straight-flight position estimate

Set `LWIRCAM_MOTION_COMPENSATION=1` in CATIA's startup environment to enable a constant-ground-velocity estimate for live LWIR shots. It is off by default. The estimate advances the reported position using ground speed, ground course (not nose heading), and the measured server interval. WGS84 local curvature converts north/east displacement to latitude/longitude. For illustration, a measured 0.32 s at 15 m/s gives 4.8 m of estimated travel; this is not a fixed delay to subtract or a claim that all shots now incur that delay.

The estimated position is written to normal GPS tags and the pose fields used for hotspot projection. EXIF also retains `original_lat_deg`, `original_lon_deg`, the measured interval and `position_compensation=constant_ground_velocity_estimate`. Attitude, altitude and AGL remain unchanged. Delays outside 0-2 seconds, speed outside 0-100 m/s, invalid inputs or latitude outside 85 degrees are not corrected. Missing timing from an older server also leaves position unchanged.

This is a partial, explicitly labeled estimate, not exposure synchronization. It assumes constant velocity and does not correct banking, acceleration, climb, or unknown buffering. Test straight passes in both directions with compensation off and on before relying on it. A smaller numerical residual on one pass is not sufficient validation. Keep original metadata for comparison.

A practical next estimate is callback arrival minus a measured camera lag, using time-aligned FC position and attitude. Interpolate bracketing poses when available; reserve bounded forward prediction for cases where later samples are missing. A 20 ms residual timing error at 15 m/s contributes 0.30 m before rotation effects, so perfect exposure timestamps are not a prerequisite if the measured total error stays inside the mission budget. See the [effective-lag model](#).

Before relying on accurate moving-aircraft GPS, validate either exposure timestamps or an effective-lag estimate and its uncertainty. Neither the lag estimator nor exposure-time pose interpolation is implemented yet. Record errors over stationary, slow and faster passes, in both directions, and across bank/pitch/yaw. Do not tune the camera mounting to hide a timing error.

Other remaining limits include GNSS bias, terrain slope, AGL uncertainty, motion blur and the local flat-ground model. A target on terrain higher than the reference plane will be projected to the wrong intersection. Choose an application error budget before deciding whether a result is acceptable.

The EXIF fields `capture_pose_synchronized=false` and `absolute_accuracy_m=unknown` are intentional. `verified: 1` does not change them.

## 12. Quick Troubleshooting

| What you see                                 | Try this first                                                                                          |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------|
| No detectable checkerboard                   | Test a matte/foil patch, improve thermal contrast, check 7x6 inner-corner count                         |
| Only front-on photos are detected            | Remove foil reflections; reduce tilt; make squares larger                                               |
| Many corners rejected near edges             | Keep the whole border visible and check focus                                                           |
| Different dimensions in one set              | Separate camera modes; never resize to force a match                                                    |
| Good score but poor ground positions         | Check mounting, image orientation, AGL and timing                                                       |
| Mounting result points the wrong way         | Inspect labels 0, 1 and 7; correct physical board axes or reverse ordering                              |
| Repeated mounting results disagree           | Improve leveling, corner contrast and board size; do not average matrix entries                         |
| <code>unverified_camera_calibration</code>   | Complete the review gate; do not merely bypass it                                                       |
| <code>invalid_camera_calibration</code>      | Check YAML syntax, required keys and exact capture dimensions                                           |
| <code>missing_shot_pose</code>               | Use a native-orientation JPEG with CATIA flight metadata, not an ordinary exported image                |
| <code>no_temperature_plane</code>            | Keep the native <code>.jpg.raw</code> companion beside the matching JPEG; brightness is not temperature |
| <code>invalid_native_frame</code>            | Restore the original complete JPEG/raw pair; do not mix frames or change dimensions                     |
| <code>no_hotspots</code> on the checkerboard | Expected if it is below detector thresholds; corner calibration is a different operation                |

# 13. Save A Reusable Calibration Pack

---

Keep one folder per camera, lens, focus and mounting combination:

```
tiny1c-aircraft-date/  
  notes.txt  
  photos/          original lens-fitting captures  
  check/           independent views  
  mount/           mounting captures and axis photographs  
  tiny1c-lens.yml  
  tiny1c-mounted.yml  
  *.report.txt  
  *-previews/  
  ground-checks.csv  
  temperature-checks.txt
```

Record camera identity, lens, focus mark, image dimensions, measured board pitch, mounting photographs, FC board alignment, software version, dates and reviewer. Record actual errors and reference uncertainty, not just a pass label.

Repeat lens calibration after changing lens, focus or resolution. Repeat mounting checks after moving the camera, changing the bracket, a hard landing or changing FC alignment. Recheck temperature after relevant sensor service or when measurements become inconsistent. Retaining the old pack makes comparison straightforward instead of starting from memory.

# References And Scope

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- [OpenCV camera calibration tutorial](#): internal corners, multiple views, camera matrix and distortion.
- [OpenCV pose estimation](#): object-to-camera pose convention.
- [CATIA geolocation implementation guide](#): metadata, calibration keys and runtime limitations.

The thermal-board construction is a practical workshop approach, not a manufacturer-certified calibration target. Photo counts and pixel-error bands are starting points for diagnosis, not guaranteed Tiny1-C performance. The illustrations are original schematics. No actual camera was calibrated while preparing this guide.